

Topics in Deep Learning Theory (Spring 2026)

FINAL PROJECT

Instructor: Lei Wu

Due: June 19, 2026

This is a preliminary draft. An updated version will be released within a week!

Important Information

- The project should be completed in a group of 1-3 people. Once the group is formed, please sign up your group in the following spreadsheet: <https://docs.qq.com/sheet/DTUNsYU1MUVVsQ2V1?tab=BB08J2>. We will use this spreadsheet to organize the presentation ordering (which will be randomly permuted).
- Important dates:
 - May 5, 11:59PM The project proposal submission
 - June 3, 5, & 12 The presentation dates
 - June 19 11:59PM The final slide submission.
- Submission instruction:
 - Please send the proposal one-page pdf and final slide to Zilin, Binghui and Yuhao (with Lei cc'd) ¹.
 - Each group only needs to send one email with subject: “TDLT2025-final-project-proposal/slide”.
 - Make sure to include the group information (group member names, student IDs, and whether each member is a graduate or undergraduate student) in your proposal and final slide.

The deadlines outlined above are tentative and may be adjusted as necessary based on the number of participating teams.

Project Task

You should select one of the following tasks to complete. No matter which task you choose, you need to give a presentation and submit the slides.

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Task 1: Literature Survey

In this project, you are required to explore a specific topic in deep learning theory. Your task is to select and delve into the chosen topic, and give a presentation to explain the details to other students and the instructor. You need to not only thoroughly review existing works but also conduct your own explorations through either numerical experiments or mathematical analysis.

A list of tentative topics is provided below for your consideration: *edge of stability*, *curriculum learning*, *length generalization*, *scaling laws*, *understanding Adam and Muon*, *in-context learning*, *transformer approximation*. You are not restricted to these topics; you are encouraged to explore and present on other topics of interest, subject to **prior approval from the instructor**.

Requirements for Task 1

Your survey, presentation and final slides should cover the following four aspects:

- **Problem Introduction:** Explain what the problem is and why it is important. Discuss its interesting and important aspects, providing context about its relevance in its field or broader applications.
- **The Essence of the Problem:** Describe the specific nature of the chosen problem and the difficulties involved in understanding and solving it. Highlight what makes this problem particularly challenging or unique.
- **Review of Existing Works:** Present an overview of the existing works related to your chosen problem. This could include both theoretical and empirical approaches. Be sure to cite your sources appropriately.
- **Further Explorations:** Share your group’s own explorations regarding the problem. This could be either new insights or further questions that your investigation has raised. Emphasize any original contributions or perspectives your group has developed.

Task 2: Predicting Loss Curve of LLM Pretraining

In this project, you are tasked with investigating the **scaling laws** in large language models (LLMs) through a series of experiments. Scaling laws have been fundamental in shaping the training of large-scale deep learning models, indicating that model performance improves according to a power law as both the size of the training data and the model itself grow. However, the underlying principles behind the emergence of these laws during complex model training remain poorly understood, especially regarding how the learning rate schedule (LRS) influences them.

In LLM pre-training, the popular LRSs include cosine schedule [Loshchilov and Hutter, 2016, Touvron et al., 2023], warmup-stable-decay (WSD) schedule [Hu et al., 2024] and multi-step schedule [Bi et al., 2024]. Specifically, consider an LRS $\boldsymbol{\eta} := (\eta_1, \eta_2, \dots, \eta_T)^\top \in \mathbb{R}_{\geq 0}^T$ over a total of T steps. We regard the loss of LLM pre-training at each step as a function of $\boldsymbol{\eta}$, i.e.,

$$L_t = \mathcal{L}_{\boldsymbol{\Theta}}(\boldsymbol{\eta}_{\leq t}), \quad \boldsymbol{\eta}_{\leq t} := (\eta_1, \eta_2, \dots, \eta_t).$$

The functional form of $\mathcal{L}_{\boldsymbol{\Theta}}$ may follow, for example, the momentum law of [Tissue et al., 2024] or the multi-power law of [Luo et al., 2024]. The parameter vector $\boldsymbol{\Theta}$ is fitted from observed training loss curves and then used to predict or extrapolate the loss under new learning-rate schedules.

Requirements for Task 2

Your experiments, presentation and final slides should cover the following three aspects:

- **Problem Introduction:** Explain what the problem is and why it is important. Discuss its interesting and important aspects, providing context about its relevance in its field or broader applications.
- **Experimental Reproduction:** Reproduce the experiments in these two papers: [Tissue et al., 2024], [Luo et al., 2024], and beyond. You can start by referring to the official code at <https://www.alphaxiv.org/abs/2408.11029> and <https://github.com/thu-yao-01-luo/MultiPowerLaw>² as well as some other GitHub repositories. For your experiments, please use the loss curves provided at <https://disk.pku.edu.cn/link/AA65601B1151C64439B656A016D0FD9A9A>. This is a temporary link and will be updated soon. Specifically, fit the model on cosine LRS and evaluate its prediction performance on WSD LRS.
- **Method Development:** Develop your own method for fitting and predicting loss curves. After reproducing the existing approaches, analyze their limitations and propose a modified or alternative fitting strategy. You should then evaluate the prediction performance of your proposed approach and compare it with the reproduced baselines. You may read some relevant literature [Li et al., 2025, Tissue et al., 2024, Zhang et al., 2026].

If you choose Task 2, you should upload all your experimental code to a GitHub repository and include the necessary documentation to ensure the code can be properly executed. Please include the link to your GitHub repository in your slides when you submit them.

Grading Policy

The final project will be graded according to the following components:

- Proposal (5pts): You are required to submit a one-page PDF proposal that clearly states the task you have selected, outlines your proposed approach, and specifies the research questions you intend to investigate. A tentative bibliography should also be included. The reference list is not included in the page limit. After the proposal submission deadline, changes to team membership will not be permitted. If necessary, you may modify your project task by submitting a revised one-page proposal and obtaining approval from both Lei and TAs via email.
- Presentation (35pts): You are required to deliver a **20-minute (to be determined)** presentation including a Q&A session (the exact duration will be determined based on the number of projects) on your project. We will make a further announcement about the presentation schedule later.
- Final slide (20pts): Please submit your polished slides. You do not need to explain all the content on your slides. Ensure that your presentation is clear, well-organized, and engaging. Use visuals, diagrams, or any relevant media to help illustrate your points. Remember to specify the division of labor among team members at the end.

²These codebases are outdated and they probably do not work well with the current version of PyTorch.

Evaluation Criteria

The assessment will be based on the following criteria:

- Clarity and depth of the problem introduction.
- Understanding and articulation of the problem's nature.
- Thoroughness, critical insight, and synthesis in the review of existing literature. (Task 1)
- Rigor, validity, solidness and reproducibility of your experimental work. (Task 2)
- Originality and insightfulness of your group's own exploration.
- Effectiveness of the presentation, including structure, visual design, and oral delivery.

References

- [Bi et al., 2024] Bi, X., Chen, D., Chen, G., Chen, S., Dai, D., Deng, C., Ding, H., Dong, K., Du, Q., Fu, Z., and Others (2024). DeepSeek LLM: Scaling open-source language models with longtermism. *arXiv preprint arXiv:2401.02954*.
- [Hu et al., 2024] Hu, S., Tu, Y., Han, X., He, C., Cui, G., Long, X., Zheng, Z., Fang, Y., Huang, Y., Zhao, W., and Others (2024). MiniCPM: Unveiling the potential of small language models with scalable training strategies. *arXiv preprint arXiv:2404.06395*.
- [Li et al., 2025] Li, B., Chen, F., Huang, Z., Wang, L., and Wu, L. (2025). Functional scaling laws in kernel regression: Loss dynamics and learning rate schedules. *arXiv preprint arXiv:2509.19189*.
- [Loshchilov and Hutter, 2016] Loshchilov, I. and Hutter, F. (2016). SGDR: Stochastic gradient descent with warm restarts. *arXiv preprint arXiv:1608.03983*.
- [Luo et al., 2024] Luo, K., Wen, H., Hu, S., Sun, Z., Liu, Z., Sun, M., Lyu, K., and Chen, W. (2024). A multi-power law for loss curve prediction across learning rate schedules. In *NeurIPS 2024 Workshop on Mathematics of Modern Machine Learning*.
- [Tissue et al., 2024] Tissue, H., Wang, V., and Wang, L. (2024). Scaling law with learning rate annealing. *arXiv preprint arXiv:2408.11029*.
- [Touvron et al., 2023] Touvron, H., Martin, L., Stone, K., Albert, P., Almahairi, A., Babaei, Y., Bashlykov, N., Batra, S., Bhargava, P., Bhosale, S., and Others (2023). LLaMA 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*.
- [Zhang et al., 2026] Zhang, H., Wen, K., and Ma, T. (2026). Configuration-to-performance scaling law with neural ansatz. *arXiv preprint arXiv:2602.10300*.